

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Computer Science and Engineering	SVNIT, Surat	7.70 (current)	2024-2028
Higher Secondary	CBSE Board	83.90%	2024
Secondary	CBSE Board	95.90%	2022

PROJECTS

- Algorithm Visualizer**
React.js, JavaScript, CSS
 - Developed an interactive web application to visualize shortest-path graph algorithms (BFS, DFS, Dijkstra's) step-by-step on a customizable 2D grid.
 - Optimized frontend rendering performance utilizing React Hooks (useState, useEffect) to manage dynamic grid states and control animation frame rates smoothly.
 - Bridged theoretical data structures with practical frontend engineering, translating complex competitive programming concepts into executable JavaScript logic.
- ElectroProtovolt Lab (AI-Powered Circuit Simulator)**
React.js, Node.js, C++, Groq API
 - Built an interactive virtual laboratory using React.js that enables students to design basic electrical circuits through a drag-and-drop interface.
 - Engineered a C++ validation engine utilizing graph traversal algorithms to analyze circuit topology, successfully detecting open loops and short circuits before execution.
 - Integrated the Groq API via a Node.js backend to process the validated circuit structures, providing students with AI-generated step-by-step mathematical solutions and conceptual explanations.
- Library Management System**
Full-Stack Database Application
 - Engineered a comprehensive platform managing a collection of 300+ books, featuring online reading capabilities and a robust issue/submission tracking system.
 - Designed a relational database schema with optimized queries for user management, book inventory, and transaction history, demonstrating strong DBMS principles.
 - Implemented automated late fee calculations based on return dates to ensure accurate financial record-keeping.

TECHNICAL SKILLS

- Computer Languages:** C/C++, Python, JavaScript, SQL, HTML, CSS
- Core Technologies:** React.js, Node.js, Express.js, MongoDB, REST APIs, Git

KEY COURSES TAKEN

- Computer Science:** Database Management System, Object-Oriented Programming, Computer Networks, Design and Analysis of Algorithms

ACHIEVEMENTS & ACTIVITIES

- JEE Mains** 2024
 - Achieved **99.52 percentile** with a National Rank of **7816** in the Common Ranking List.
- Competitive Programming**
 - Actively training for ICPC using C++. Achieved a peak rating of **1236 on Codeforces** and consistently solve advanced algorithmic challenges on LeetCode.
- Extracurricular Interests**
 - Reading, Music, Tennis, Mathematics Enthusiast.